ZAENAL SYAMSYUL ARIEF

AI Engineer | Machine Learning Engineer | Data Science
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PROFESSIONAL SUMMARY

8th-semester Informatics Engineering student (GPA 3.92/4.00) with hands-on experience across the AI engineering pipeline: SQL and API-based data extraction, data cleaning, preprocessing and Exploratory Data Analysis (EDA), and end-to-end Machine Learning and Deep Learning model development (supervised and unsupervised learning, CNN, NLP/Transformer-based architectures). Practical experience integrating Large Language Models (LLM) into applied products including LLM-powered chatbots and Google Vertex AI integration and building automated data pipelines in production-style internship and research settings. Comfortable working independently or in cross-functional teams to translate business needs into working AI solutions.

EXPERIENCE & KEY PROJECTS

MyCareAI - FER, Chatbot & Adaptive Persona Integration — *Feb 2026 – Jun 2026*
Ongoing Applied Research / Thesis Project

- Designed the end-to-end camera-to-model-to-chatbot integration flow, combining a CNN/VGG16-based five-class Facial Emotion Recognition (FER) pipeline with LLM-driven chatbot responses.
- Built adaptive persona logic to reconcile emotion-text mismatches with contextual, rule-based responses direct experience translating a business requirement (mental-health monitoring) into a working AI system.
- Trained and evaluated deep learning models (CNN/VGG16) on FER-2013 and RAF-DB datasets, covering model training, evaluation, and iterative tuning.

Tech Stack: Python, TensorFlow, Keras, VGG16 (CNN), NLP, LLM/Chatbot Integration, Adaptive Persona Logic

Data Science Intern - NoLimit Indonesia — *Jul 2025 – Nov 2025*

- Built automated ETL pipelines to extract, transform, and process real-time social media data from multiple platforms using Python and Elasticsearch directly applying SQL/API/structured-data extraction and data cleaning skills.
- Developed and deployed a supervised-learning sentiment classification model achieving high accuracy on public social data, including preprocessing, feature engineering, and evaluation.
- Built internal Python-based tools to speed up data analysis and improve cross-division collaboration.
- Collaborated in a cross-functional data science team to deliver shared analytical objectives.

Machine Learning Cohort - Coding Camp powered by DBS Foundation — *Feb 2025 – Jul 2025*

- Developed end-to-end ML models with Python, TensorFlow, and Scikit-learn across supervised learning (classification), covering the full workflow from EDA and preprocessing to model training and evaluation.
- Applied deep learning to NLP-based text sentiment analysis and CNN-based image classification.
- Designed reliable ETL pipelines to keep data integrity and workflow efficiency, including structured data extraction and feature engineering.
- Built an unsupervised-learning-informed recommendation system using content-based, collaborative filtering, and hybrid approaches.

Capstone Project: MyCareAI - Emotion-Detection Web App with Chatbot — *Jun 2025*

- Built an NLP-based emotion classification model as the core of a Dual-AI architecture, integrating Google Vertex AI (LLM) with a custom classification model via API hands-on Generative AI / LLM-integration experience.
- Reached 92% accuracy across four emotion categories, improving the relevance of chatbot responses.
- Integrated the trained model into an interactive web-based chatbot, exposing the model to downstream services.

Tech Stack: Python, TensorFlow, Scikit-learn, NLP, Google Vertex AI (LLM Integration)

StuntLytics - AI-Based Stunting Risk Prediction for Local Government — *Oct 2025*
Team CiptaMuda AI (APHacton 2025) — Data Preprocessing & Systems Integration Specialist

- Built a 100,000-record synthetic dataset from BPS and SSGI data, including data cleaning and feature engineering.
- Developed a supervised Random Forest prediction model with a Scikit-learn preprocessing pipeline, covering model training and evaluation.
- Integrated the model into an interactive Streamlit dashboard backed by Elasticsearch, and shipped an LLM-powered chatbot feature ("InsightNow") for automated policy-recommendation insights.

Tech Stack: Python, Scikit-learn, Streamlit, Elasticsearch, Random Forest, LLM Integration

SKILLS

Programming & Data Engineering: Python (primary language for AI/ML development), SQL, data extraction & collection from databases, APIs, and CSV/structured sources, Elasticsearch, PostgreSQL, ETL pipeline design, web scraping.

Data Preparation & EDA: Data cleaning, preprocessing, feature engineering, statistical analysis, and Exploratory Data Analysis (EDA) across multiple applied projects.

Machine Learning: Supervised learning (classification, Random Forest) and unsupervised learning (collaborative/content-based filtering), model training & evaluation, hands-on with Scikit-learn.

Deep Learning: CNN (VGG16-based FER pipeline), NLP/Transformer-based sentiment and emotion classification, TensorFlow & Keras.

Generative AI / LLM: LLM API integration (Google Vertex AI), prompt-driven chatbot design, Dual-AI architecture combining LLMs with custom classifiers.

Tools & Practices: Git version control, Jupyter/Google Colaboratory, Google Cloud, model deployment, cross-functional team collaboration.

Soft Skills: Problem solving, analytical thinking, communication, collaboration, time management, adaptability.

EDUCATION

Institut Teknologi Garut — Bachelor of Informatics Engineering *Sep 2022 – Present*
GPA: 3.92 / 4.00

CERTIFICATIONS

Multiple certifications from Dicoding Indonesia covering Machine Learning, Data Processing, and Data Analysis.
Full list: Bitly | bit.ly/40ANaNH